

KW 10: Financial Markets

Ch 4, Ch 5.2, Ch 5.3 + Lecture 3

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

LECTURE 3 CONTEXT: FINANCIAL MARKET OVERVIEW

The financial market plays a central role in an economy. It is complementary to the goods market (covered in Lecture 2). A financial market is a generic term for any market in which trading of financial instruments takes place.

Financial Market Taxonomy

- **Money market:** Short-term money supply and demand
- **Bonds market** (also credit market): Financial assets and liabilities
- **Capital market:** Medium and long-term capital requirements and supply

Simplifying assumption for this course: only two forms of assets exist - money and bonds. This allows us to understand how the interest rate is set and the role of the central bank.

- **Money:** Can be used for transactions, but does not earn interest
- **Bonds:** Yield an interest rate i , but cannot be used for transactions
- Individuals choose what fraction of their wealth to hold as money vs. bonds

Learning Objectives (Lecture 3)

- The role of money in the national economy
- Money supply and demand
- Monetary policy through central banks
- Financial market for a model of the economy in the short run
- Derivation of the LM curve for the IS-LM model
- Literature: Blanchard, Chapters 4 and 5.2

What Is Money?

- Three classical functions (from Microeconomics): medium of exchange, unit of account, means of storing value
- Historically many forms: stones, shells, gold, receipts, cigarettes, Bitcoin, ...
- Inflation affects the functioning of money (covered in Lecture 6)

Types of Money in a Modern Economy

Central banks measure money supply using monetary aggregates. Definitions used by the Swiss National Bank (SNB):

- **Monetary base (M0):** Banknotes in circulation + sight deposits of domestic commercial banks held at the SNB
- **M1:** Currency + sight deposits in Swiss francs from residents
- **M2:** M1 + savings deposits held at banks (excluding pillar 2/3a) in Swiss francs
- **M3:** M2 + time deposits held at banks in Swiss francs

SNB Money Aggregates (1990-2025)

All three aggregates grew substantially since 1990. M3 (broadest) reached ~1,100B CHF by 2025. M1 spiked around 2020 (pandemic monetary expansion), peaking near 800B CHF before declining. Monetary base is the smallest.

CHAPTER 4: FINANCIAL MARKETS I

Financial markets determine the cost of funds for firms, households, and the government. The chapter simplifies to two financial assets: money (pays no interest) and bonds (pay interest rate i). Focus is on how the interest rate is determined and the role of the central bank.

Semantic Traps: Money, Income, and Wealth

Money: Currency + checkable deposits. Used for transactions. A STOCK variable.

Income: What you earn (wages + interest + dividends). A FLOW variable.

Saving: Part of after-tax income not spent. A FLOW.

Financial wealth: All financial assets minus liabilities. A STOCK.

Investment: Purchase of NEW capital goods (NOT financial assets).

Financial investment: Purchase of shares/bonds.

4-1 The Demand for Money

The Portfolio Choice: Money vs. Bonds

- **Money:** Can be used for transactions, pays no interest. Two types: currency (coins/bills) and checkable deposits (bank accounts with checks/debit cards)
- **Bonds:** Pay positive interest rate i , but cannot be used directly for transactions. Transaction costs exist for buying/selling bonds.
- You should hold BOTH: enough money for transactions, rest in bonds for interest

Two Key Determinants

- **(1) Level of transactions:** More transactions $\rightarrow$ need more money on hand. Roughly proportional to nominal income ($\$Y$).
- **(2) Interest rate on bonds:** Higher $i \rightarrow$ more willing to deal with inconvenience of selling bonds $\rightarrow$ hold less money. At very high rates, minimize money holdings.

The Money Demand Equation

$$M^d = \$Y \cdot L(i) \quad (-)$$

$$M^d = \text{demand for money}$$

$$\$Y = \text{nominal income}$$

$$L(i) = \text{decreasing function of interest rate}$$

- Money demand increases **proportionally** to nominal income

- Money demand depends **negatively** on the interest rate
- What matters is NOMINAL income (not real) - if prices double, need twice as much money for same transactions

Figure 4-1: Money Demand Curve

- Downward sloping: lower interest rate -> more money demanded
- An increase in nominal income shifts the entire curve RIGHT (more money demanded at every interest rate)
- At a given interest rate i , an increase in nominal income shifts the money demand curve to the RIGHT
- Points along the curve: moving from low i to high i , money demand DECREASES (liquidity preference falls)

Empirical Validation: Swiss Cash-Deposit Ratio (Lecture)

The lecture presents Swiss empirical data (1990-2025) showing the cash-holding coefficient alongside the return on 10-year Swiss government bonds:

$$L(\hat{i}) = \frac{M^d}{PY} \quad (\text{cash-deposit ratio})$$

- As the 10Y bond return fell from ~6% (early 1990s) to near 0% (2015-2020), the cash-holding coefficient INCREASED from ~30% to over 100%
- This is consistent with the theory: lower interest rates -> higher money demand relative to income
- When bonds pay almost nothing, the opportunity cost of holding money vanishes

Who Holds U.S. Currency?

Total U.S. currency in circulation: \$750 billion (2006)

U.S. households held only \$170 billion

~\$500 billion (66%) held by FOREIGNERS

Countries like Russia (\$80B), Argentina (\$50B+) hold dollars as safe assets

Result: The rest of the world makes an interest-free loan to the U.S.

4-2 Determining the Interest Rate: I

Assumes only money is currency (central bank money). Checkable deposits introduced in 4-3.

Equilibrium Condition

$$M = \$Y \cdot L(i)$$

- The interest rate i must be such that people are willing to hold exactly the existing money supply M

Figure 4-2: Interest Rate Determination

- Money demand curve: downward-sloping
- Money supply curve: vertical line at M
- Equilibrium at intersection point A

Effects of Changes in Nominal Income (Figure 4-3)

- Increase in nominal income -> money demand shifts RIGHT -> interest rate INCREASES

- Reason: at old rate, money demand > supply. Higher i reduces demand back to equilibrium

Effects of Increase in Money Supply (Figure 4-4)

- Increase in $M \rightarrow$ money supply shifts RIGHT $\rightarrow$ interest rate DECREASES
- Reason: lower i increases demand to match the now larger supply

Monetary Policy and Open Market Operations

- Central bank changes money supply by **buying or selling bonds in the bond market**
- **Open market operations:** transactions in the 'open market' for bonds

Open Market Operations

EXPANSIONARY OMO: Central bank BUYS bonds, pays with newly created money

$\rightarrow$ Money supply increases $\rightarrow$ Bond prices increase $\rightarrow$ Interest rate DECREASES

CONTRACTIONARY OMO: Central bank SELLS bonds, removes money from circulation

$\rightarrow$ Money supply decreases $\rightarrow$ Bond prices decrease $\rightarrow$ Interest rate INCREASES

Central Bank Balance Sheet (Figure 4-5)

- Assets: Bonds held by the central bank
- Liabilities: Money (currency) in the economy
- OMOs lead to equal changes in assets and liabilities

Bond Prices and Bond Yields

Consider a one-year bond (T-bill) with face value \$100:

$$i = \frac{\$100 - \$P_B}{\$P_B}$$

$$\$P_B = \frac{\$100}{1 + i}$$

- Higher bond price $\rightarrow$ lower interest rate (and vice versa)
- 'Bond markets went up' = bond prices rose = interest rates fell

Interest Rates in Practice (Lecture)

Global context: Interest rates on 10-year government bonds fell worldwide for decades (USA, Germany, Japan, Switzerland). From peaks of 10-16% in the early 1980s, rates declined to near zero or even negative by 2020.

- **Zero lower bound:** How negative can an interest rate be?
 - > Cash pays zero interest, but holding it causes costs (storage, insurance, theft risk)
 - > In practice, rates went slightly negative in Europe and Japan (investors accepted small losses for safety)

Choosing Money or Choosing the Interest Rate?

The lecture presents these as two distinct equilibrium regimes with equal importance:

Regime 1: Money Supply Control

- The central bank provides money supply M , so that $M = M_S$
- The interest rate is determined endogenously in equilibrium

$$M^S = M^d \iff M^S = PY \cdot L(i)$$

- Graph: vertical M_S line intersects downward-sloping M_d curve at equilibrium point A

Regime 2: Interest Rate Control

- The central bank sets the interest rate to i_0
- The money supply is determined endogenously: the central bank provides as much money as demanded at i_0

$$M^d(i_0) = M^S \quad (\text{money supply adjusts to demand at } i_0)$$

- Graph: horizontal line at i_0 intersects downward-sloping M_d curve at equilibrium point A

Expansionary Monetary Policy: Both Regimes (Lecture)

The lecture shows both regimes side by side for an expansionary policy:

Under Money Supply Control

- Central bank increases money supply from M_1 to M_2 (supply curve shifts right)
- Equilibrium interest rate falls from i_1 to i_2
- Money demand rises accordingly
- The interest rate is determined **endogenously**

Under Interest Rate Control

- Central bank aims for a lower interest rate: reduces i from i_1 to i_2
- At the lower rate, money demand increases from M_1 to M_2
- Central bank increases money supply to M_2 to satisfy demand
- The money supply is determined **endogenously**

Equivalence of the Two Descriptions

Money supply control and interest rate control produce the SAME outcome.

Modern central banks think in terms of choosing the interest rate.

News says: 'The Fed decided to increase the interest rate'

(not 'decrease the money supply').

Historically: money supply targeting (late 1970s to early 1990s),

then shift to interest rate targeting (since the early 1990s),

with QE revival of money supply operations for long-term bonds.

4-3 Determining the Interest Rate: II (With Banks)

In reality, money includes currency AND checkable deposits. Checkable deposits are supplied by private banks, not the central bank.

What Banks Do

- Banks are **financial intermediaries** - receive funds and use them to buy assets or make loans
- Special because their liabilities are money (checkable deposits)

Bank Balance Sheet (Figure 4-6)

Assets	Liabilities
Reserves	Checkable deposits
Loans (~70%)	
Bonds (~30%)	

Reserves

- Banks keep some funds as reserves (cash + deposits at central bank)
- Three reasons: (1) daily cash flow mismatches, (2) interbank check clearing, (3) **reserve requirements**
- U.S. reserve requirement: at least **10% of checkable deposits**

Central Bank Money (The Monetary Base)

$$H = CU + R \quad (\text{Central Bank Money} = \text{Currency} + \text{Reserves})$$

Demand for Central Bank Money

Simplifying assumption: people hold only checkable deposits (no currency).

- Demand for checkable deposits = demand for money by people:

$$M^d = \$Y \cdot L(i)$$

Demand for reserves by banks (theta = reserve ratio):

$$H^d = \theta \cdot M^d = \theta \cdot \$Y \cdot L(i)$$

Equilibrium

$$H = \theta \cdot \$Y \cdot L(i)$$

- H = supply of central bank money (controlled by central bank via OMOs)
- Same qualitative results as before: increase in H -> lower interest rate

Federal Funds Market and Federal Funds Rate

In the U.S., banks trade reserves in the FEDERAL FUNDS MARKET.

The interest rate in this market = FEDERAL FUNDS RATE.

This is the main indicator of U.S. monetary policy.

Changes in the federal funds rate make front-page news.

4-4 The Liquidity Trap

The Zero Lower Bound

- The interest rate on bonds **cannot be negative** -> the **zero lower bound (ZLB)**
- When $i = 0$, monetary policy cannot decrease it further -> **liquidity trap**

Why the Zero Lower Bound Exists

- At $i = 0$, both money and bonds pay the same interest rate: zero
- People are **indifferent** between holding money or bonds
- Additional money supply is simply absorbed without lowering rates

Figure 4-8: The Liquidity Trap

- Money demand curve becomes **horizontal at $i = 0$**
- If money supply is at/beyond the flat portion: interest rate = 0
- Further increases in money supply have NO effect on interest rate
- Monetary policy is **ineffective** in the liquidity trap

The Liquidity Trap in Practice

During the 2008 crisis:

- Bank of England cut rate from 5% to 0.5%
- Bank reserves expanded from 23.6B to 265.5B pounds
- Rate remained at 0.5% despite massive money expansion
- This is exactly what liquidity trap theory predicts

Money Creation by Commercial Banks (Lecture Detail)

The lecture uses the following notation for money creation:

M = money supply, H = central bank money (high-powered money, $M0$)

CU = cash in circulation, D = deposits of non-banks

R = reserves, θ = reserve ratio, b = cash proportion

In an economy with commercial banks, the central bank can only control the monetary base H directly. The money supply M is controlled indirectly.

Why Banks Hold Reserves

- Deposits and withdrawals are not equal - banks must hold cash on hand
- To cover debts to other banks
- To fulfill legal **reserve requirements**
- Reserve ratio: banks hold fraction θ of deposits as reserves

$$R = \theta \cdot D$$

The Money Creation Multiplier (Geometric Series)

The monetary base H consists of cash in circulation plus reserves:

$$H = CU + R \iff H = CU + \theta D$$

Cash and deposits are fixed proportions b and (1-b) of total money M:

$$CU = b \cdot M, \quad D = (1 - b) \cdot M$$

If the central bank increases H by one unit, how does this affect M?

- The commercial bank can lend a fraction (1-theta) of the deposit
- Of that loan, fraction (1-b) flows back into the banking system as new deposits
- This process continues indefinitely, yielding a geometric series:

$$1 + (1 - \theta)(1 - b) + [(1 - \theta)(1 - b)]^2 + \dots = \frac{1}{1 - (1 - \theta)(1 - b)}$$

The money creation multiplier result:

$$M = \frac{1}{b + \theta(1 - b)} \cdot H$$

- The multiplier is always > 1 (since $b < 1$ and $\theta < 1$)

Textbook Notation (Blanchard)

Blanchard uses c instead of b for the cash proportion. The formulas are equivalent:

$$CU^d = c \cdot M^d, \quad D^d = (1 - c) \cdot M^d, \quad R^d = \theta(1 - c) \cdot M^d$$

Demand for central bank money:

$$H^d = [c + \theta(1 - c)] \cdot \$Y \cdot L(i)$$

Equilibrium:

$$H = [c + \theta(1 - c)] \cdot \$Y \cdot L(i)$$

Money multiplier (textbook):

$$M = \frac{H}{c + \theta(1 - c)} = \text{Money Multiplier} \times H$$

$$\text{Money multiplier} = \frac{1}{c + \theta(1 - c)} > 1$$

- Example:

$$c = 0.4, \theta = 0.1 \implies \text{multiplier} = \frac{1}{0.46} = 2.2$$

- Each dollar of central bank money leads to \$2.20 of total money

Central Bank Control Channels (Lecture)

With commercial banks, the central bank controls money supply through three channels:

- **Policy rate** (0.00% in Jan 2026): The price at which commercial banks can obtain central bank money
- **Deposit rate** (0.00% in Jan 2026): Interest on deposits by commercial banks with the central bank
- **Minimum reserve rate** (4.00% in Jan 2026): Compulsory reserve balances of commercial banks with the central bank

The central bank indirectly influences the volume of lending by commercial banks and thus the amount of their deposits and the money supply M.

SNB Monetary Policy in Practice (Lecture)

In October 2022, the SNB had to intervene directly in the market because the SARON (Swiss Average Rate Overnight) differed from the SNB policy rate. The SARON is the actual market rate at which banks lend to each other overnight. When SARON diverges from the policy rate, the central bank must act to restore the transmission mechanism of monetary policy.

CHAPTER 5, SECTION 5-2: Financial Markets and the LM Relation

Why the IS-LM Model? (Lecture)

Repetition from Lecture 2: the goal is a simple model of the economy in the short run.

- In the last lecture: balance on the goods market (implicit IS curve)

$$Y = c_0 + c_1(Y - T) + I(Y, i) + G$$

- On the basis of today's lecture we can add i as an equilibrium result from the financial market
- IS-LM model in the diagram with i and Y on the axes
- Equilibrium in the financial market in the diagram: the **LM curve**
- 'LM' stands for **liquidity** and **money supply**

The LM Relation

From Chapter 4, the interest rate is determined by money market equilibrium:

$$M = PY \cdot L(i)$$

Dividing both sides by the price level P to get real terms:

$$\frac{M}{P} = Y \cdot L(i)$$

- **Real money supply** = money stock in terms of goods:

$$\text{Real money supply} = \frac{M}{P}$$

- **Real money demand** depends on real income Y and interest rate i
- Note: in the short term, the price level P is fixed
- If the money market is in equilibrium, the bonds market will be as well (Walras' law)

Deriving the LM Curve: Two Regimes

The lecture derives the LM curve under both regimes with equal emphasis:

Regime 1: Interest Rate Control (Modern, Horizontal LM)

- Central bank sets the interest rate to i_0
- The central bank endogenously adjusts the supply of money to the demand for money at that rate
- If income rises from Y_1 to Y_2 , money demand increases
- The central bank increases money supply from M_1 to M_2 to maintain i_0

- Result: the LM curve is a **horizontal line** at i_0

LM curve (interest rate control): $i = \bar{i}_0$

- Graph: left panel shows money market (M_S shifts right to match higher M_d), right panel shows flat LM line with points A (Y_1) and B (Y_2) both at i_0

Regime 2: Money Supply Control (Traditional, Upward-Sloping LM)

- Central bank keeps money supply M_S constant
- If income rises from Y to Y' , money demand increases (curve shifts right)
- With constant M_S , the interest rate must INCREASE from i_A to i_B to restore equilibrium
- Result: the LM curve is **upward-sloping** - higher income requires higher interest rate

LM curve (money supply control): $\frac{M}{P} = Y \cdot L(i)$

- Graph: left panel shows money market (M_d shifts right along fixed M_S), right panel shows upward-sloping LM curve with points A (Y, i_A) and B (Y', i_B)

The LM Curve: Summary (Lecture Slide 30)

The textbook (usually) assumes the central bank controls interest rates, and therefore uses a flat LM curve.

Summary of positions relative to the LM curve:

- ABOVE the LM curve: excess supply on the money market
- BELOW the LM curve: excess demand on the money market
- ON the LM curve: equilibrium on the money market

The bonds market is the mirror image of the money market.

The LM curve also describes the balance on the bonds market.

CHAPTER 5, SECTION 5-3: Putting the IS and LM Relations Together

The IS-LM Model

Two relations that jointly determine output and the interest rate:

$$\text{IS: } Y = C(Y - T) + I(Y, i) + G$$

$$\text{LM: } i = \bar{i}$$

Figure 5-5: The IS-LM Model

- IS curve: downward sloping (higher $i \rightarrow$ lower Y)
- LM curve: horizontal line at $i\text{-bar}$
- Equilibrium at intersection (point A): both goods market and financial markets are in equilibrium

Fiscal Policy in the IS-LM Model

An Increase in Taxes (Figure 5-6)

- Higher $T \rightarrow$ lower disposable income $\rightarrow$ lower consumption $\rightarrow$ lower demand $\rightarrow$ lower output
- IS curve shifts LEFT (from IS to IS')
- LM curve unchanged (central bank keeps $i\text{-bar}$)
- Result: Output DECREASES, interest rate UNCHANGED
- Similarly: increase in G shifts IS RIGHT $\rightarrow$ output increases, rate unchanged

Monetary Policy in the IS-LM Model

An Increase in the Interest Rate (Figure 5-7)

- Central bank raises rate from $i\text{-bar}$ to $i\text{-bar}'$
- IS curve does NOT shift
- LM curve shifts UP (from LM to LM')
- Economy moves along the IS curve to new equilibrium
- Result: Output DECREASES, interest rate INCREASES
- Mechanism: higher $i \rightarrow$ lower investment $\rightarrow$ lower demand $\rightarrow$ lower output (multiplier effect)

Policy Mix: Summary of Effects

Policy Change	IS	LM	i	Y
Increase T	LEFT	Same	Same	DOWN

Decrease T	RIGHT	Same	Same	UP	
Increase G	RIGHT	Same	Same	UP	
Decrease G	LEFT	Same	Same	DOWN	
Raise $i\text{-bar}$ (tight)	Same	UP	UP	DOWN	
Lower $i\text{-bar}$ (loose)	Same	DOWN	DOWN	UP	

Both fiscal policy (G, T shifting the IS curve) and monetary policy ($i\text{-bar}$ shifting the LM curve) can affect output. The combination of these two policies - the policy mix - determines the overall outcome.

SUMMARY OF KEY EQUATIONS AND TERMS

Core Equations

$$(4.1) \quad M^d = PY \cdot L(i) \quad \text{Money demand}$$

$$(4.2) \quad M = PY \cdot L(i) \quad \text{Money market equilibrium}$$

$$(4.4) \quad H^d = \theta \cdot PY \cdot L(i) \quad \text{Reserves demand (no currency)}$$

$$(4.6) \quad H = \theta \cdot PY \cdot L(i) \quad \text{Central bank money equilibrium}$$

$$i = \frac{100 - P_B}{P_B} \quad \text{Interest rate from bond price}$$

$$P_B = \frac{100}{1+i} \quad \text{Bond price from interest rate}$$

$$R = \theta \cdot D \quad \text{Reserves} = \text{reserve ratio} \times \text{deposits}$$

$$H = CU + R \quad \text{Monetary base} = \text{cash} + \text{reserves}$$

Money Creation

$$(\text{Lecture}) \quad M = \frac{1}{b + \theta(1-b)} \cdot H \quad \text{Money multiplier (lecture notation)}$$

$$(\text{Blanchard}) \quad M = \frac{H}{c + \theta(1-c)} \quad \text{Money multiplier (textbook notation)}$$

$$(4.A9) \quad H = [c + \theta(1-c)] \cdot \$Y \cdot L(i) \quad \text{General equilibrium}$$

IS-LM Model

$$(5.2) \quad Y = C(Y - T) + I(Y, i) + G \quad \text{IS relation}$$

$$(5.3) \quad \frac{M}{P} = Y \cdot L(i) \quad \text{LM relation (real terms)}$$

$$(5.4) \quad i = \bar{i} \quad \text{LM curve (interest rate control, horizontal)}$$

$$\text{LM: } \frac{M}{P} = Y \cdot L(i) \quad (\text{money supply control, upward-sloping})$$

Key Terms

- **Federal Reserve (Fed):** The U.S. central bank
- **Swiss National Bank (SNB):** The Swiss central bank; uses SARON as key market rate
- **Open Market Operation (OMO):** Central bank buying/selling bonds to change money supply
- **Expansionary OMO:** Buy bonds -> increase M -> decrease i
- **Contractionary OMO:** Sell bonds -> decrease M -> increase i
- **Quantitative Easing (QE):** OMOs for long-term bonds; revival since 2008
- **Forward guidance:** Central bank expectation management regarding future policy
- **Treasury bill (T-bill):** Government bond with maturity < 1 year
- **Central bank money (H):** Currency + Reserves (= Monetary Base = High-powered money = M0)
- **Reserve ratio (theta):** Reserves / Checkable deposits (min 10% in U.S., 4% SNB)
- **SARON:** Swiss Average Rate Overnight; actual interbank lending rate in Switzerland
- **Federal funds rate:** Interest rate in the federal funds market; main U.S. policy indicator
- **Zero lower bound (ZLB):** Interest rates cannot go (much) below zero
- **Liquidity trap:** At $i=0$, more money has no effect on interest rate
- **IS curve:** Downward-sloping: all (i, Y) pairs where goods market is in equilibrium
- **LM curve (modern):** Horizontal line at $i\text{-bar}$; financial market equilibrium under interest rate control
- **LM curve (traditional):** Upward-sloping; financial market equilibrium under money supply control
- **Money multiplier:** Total money as multiple of central bank money: $1/(c + \theta(1-c))$
- **Real money supply:** Money stock measured in terms of goods: M/P
- **Policy mix:** Combination of fiscal and monetary policy
- **Monetary aggregates:** M0 (base), M1, M2, M3 - progressively broader definitions of money
- **Money supply control:** Central bank sets M, interest rate adjusts endogenously
- **Interest rate control:** Central bank sets i, money supply adjusts endogenously

Key formulas for quick reference:

$$\text{Money multiplier} = \frac{1}{c + \theta(1 - c)} > 1$$

$$\text{Real money supply} = \frac{M}{P}$$

Bond price $\uparrow \iff$ Interest rate $\downarrow$